

7 CLEANUP PROCEDURES

7.1 CLEANUP PROCEDURES

This chapter describes procedures for cleaning the seal checker components. The method of cleaning should be adapted to product characteristics.



- **Before starting to clean, operators should be sure to turn off the main power switch.**
The person performing the work should lock the main power switch and keep the key in his possession while performing the work.
Injury or electrical shock may result if the equipment is turned on by another person while the work is being performed.
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- **Only wash components specifically designated in this manual.**
Keep other components dry and clean by wiping them with a dry cloth.
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7.2 Cleaning and Sterilization

This system is designed to remove the units and parts that directly contact with products for cleaning and sterilization. Always remove the detachable units and parts from the system for cleaning and sterilization. For removal and mounting procedures of such units and parts, refer to " 7.4 Cleaning and Sterilization of Each Part ".

7.3 Cleaning and Sterilization Methods

Table 7-1 shows the standard cleaning and sterilization methods recommended for this system. Actual cleaning and sterilization methods may differ depending on the nature of product, process to use the system, and microbiological generation status.

Prepare the cleaning and sterilization manual referring to Table 7-1 and carry out the daily cleaning and sterilization.

For details of each part, carefully read " 7.4 Cleaning and Sterilization of Each Part ".

NOTE

- The cleaning and sterilization methods differ depending on the nature of product, process to use the system and microbiological generation status.
For detailed information, consult the distributor you purchased or our customers' window.
- After cleaning and sterilization, thoroughly wash with water and dry.
- If spot welding affects microbiological generation status, consult the distributor you purchased or our customers' window.

Table 7-1 Cleaning/Sterilization Method

Unit, Part		Cleaning Method	Sterilization Method
<Detachable unit> • J-belt conveyor • Reject conveyor <No undetachable unit> • Remote control unit • Printer unit (Option) • Infeed conveyor unit • Seal check device	PB spec.6 (IP40)	1. Wipe cleaning A method to wipe off with cloth in which a designated cleaning solvent (neutral detergent) is impregnated followed by wiping with wet cloth. NOTE Only wipe cleaning is applied to remote control unit and printer unit.	1. Wiping sterilization A method to wipe with cloth in which disinfectant is impregnated. *Disinfectant Alcohol: 80vol% NOTE • Only wiping sterilization with alcohol: 80vol% is applied to remote control unit and printer unit. • After sterilization, wash with water and dry.

7.4 Cleaning and Sterilization of Each Part

7.4.1 RCU Cleaning

**CAUTION**

- Do not clean the operation panel of the remote control unit by rubbing it strongly or using adhesive tape.
- Do not clean the operation panel of the remote control unit with thinner, benzene, or other organic solvents.
- Do not apply detergent directly to the operation panel of the remote control unit.

1. Wash and sterilize the remote control unit, then wipe it with a cloth moistened with water.
Washing : by wiping (neutral detergent)
Sterilizing : by wiping (Alcohol: 80%)

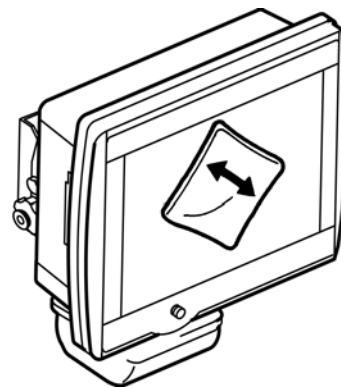


Fig.7-1 Remote Control Unit Cleaning

7.4.2 Cleaning of Conveyor Belt

- Cleaning method
1. Unlock the lever as shown in the right figure and slide the cover to the discharge side.
 2. Gently wipe the conveyor belt with a soft cloth containing neutral detergent.

**CAUTION**

**Do not lift the upper seal check conveyor by hands.
Your finger may be caught in the UP/DOWN arm.**

NOTE

Blow compressed air in the gap between the lower seal check conveyor and the upper seal check conveyor or between the infeed conveyor and the conditioning conveyor to remove dust.

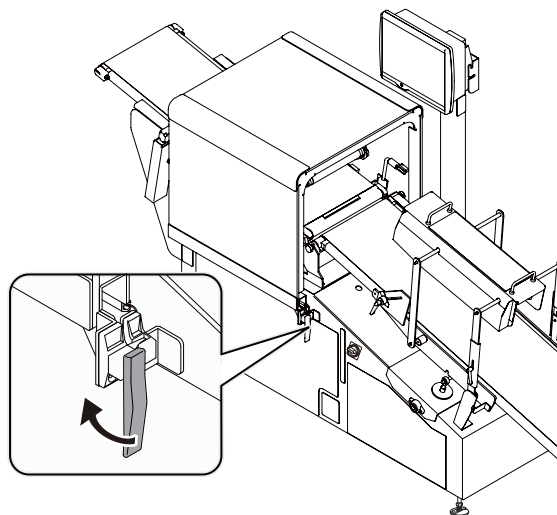


Fig.7-2 Belt Conveyor Cleaning View

- Conveyor belt ON/OFF method

For the J-belt conveyor and reject conveyor, remove the belt in the following procedure.

NOTE

- Figure shows the reject conveyor.
For the J-belt conveyor, remove the belt in the similar procedure.

1. Unscrew the bolts fixing the two cover under the conveyor and remove them.

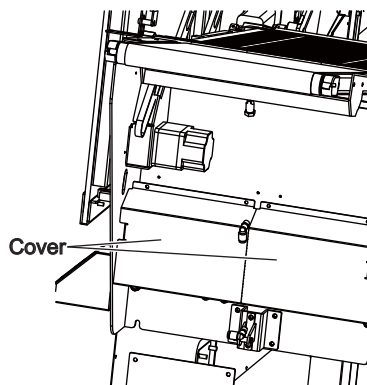


Fig.7-3 Remove the cover under the conveyor

2. Disconnect the harness connector.

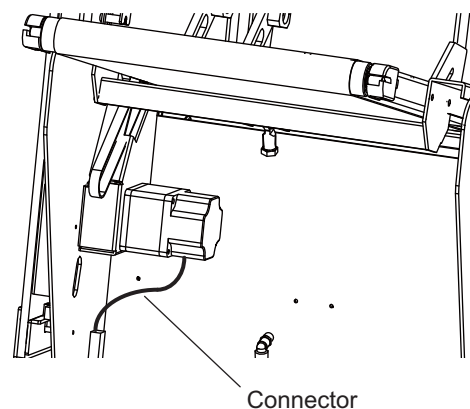


Fig.7-4 Disconnect the harness connector

3. Remove the air rejector.
4. Unscrew the bolts to remove the cover.

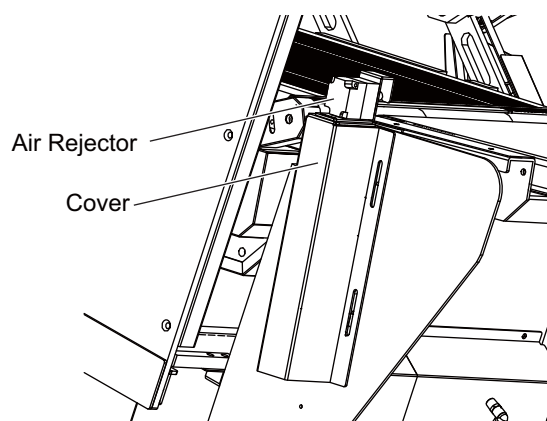


Fig.7-5 Remove the conveyor

4. Unscrew the bolts fixing the motor and remove the motor.

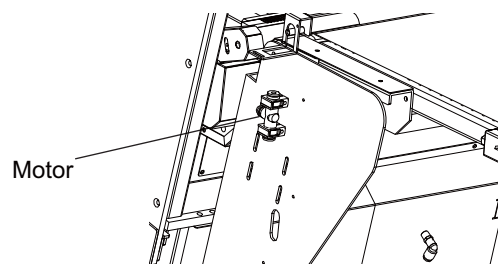


Fig.7-6 Remove the conveyor

5. Unscrew the bolts and remove the air cylinder.

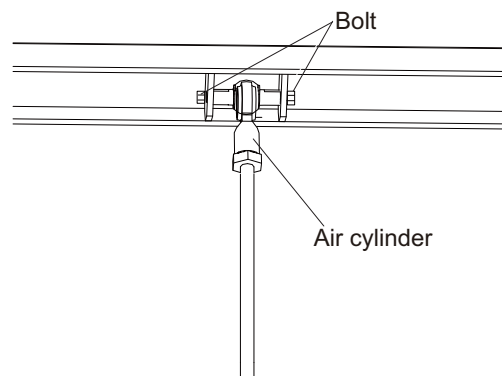


Fig.7-7 Remove the air cylinder

6. Unscrew the bolt fixing the conveyor to remove the conveyor.
7. Unscrew the bolt and remove the conveyor bracket.

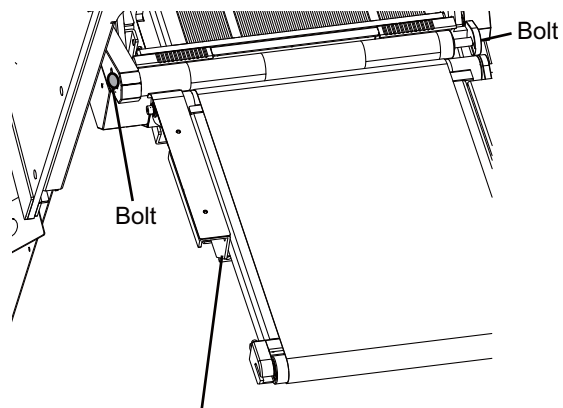


Fig.7-8 Remove the conveyor and bracket

8. Unscrew the bolt fixing the timing belt, and then remove the timing belt.

NOTE

- Figure shows the reject conveyor.
- Check that there is no scratch or cut on the timing belt. If there is any scratch or cut, contact ISHIDA service Center.



Fig.7-9 Remove the timing belt

9. To attach the conveyor, take the reverse order of removal.
- After completion of attachment, manually turn the conveyor belt to check if the timing belt is properly attached to the gear of driving motor.

7.4.3 Photo Sensor



- **Do not use organic solvents for cleaning.**
They may deteriorate the resin surface and result in the malfunction of photo sensor.

Wipe the photo sensor with a soft cloth moistened with neutral cleanser.

7.4.4 Cleaning of Cleaning Brush

The Cleaning Brush cleans on the conveyor belt during the operation. The brushes are attached to two parts, the infeed conveyor and the seal check conveyor.

7.4.4.1 Infeed Conveyor (For the specification of all-in-one height-adjustable conveyor)

Remove and clean the brush once a week or when the brush is extremely dirty.

1. Turn off the main power of equipment.
2. Check that the conveyor belt does not operate and remove the cleaning brush.



Fig.7-10 Remove the cleaning brush

3. Use neutral detergent to wash off the dirt.
4. Naturally dry the cleaning brush and attach it. Check that the tip of cleaning brush touches the belt. If not, adjust the height of attachment.



Fig.7-11 Attach the cleaning brush

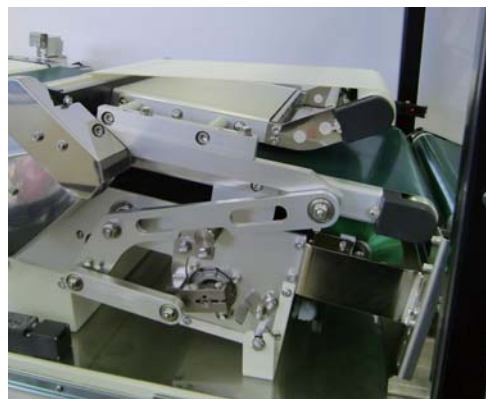
< Height adjustment of cleaning brush >

1. Loosen the height adjustment screws of cleaning brush.
2. Adjust the height so that the tip of brush touches the belt. Then loosen the screws to secure the position.

**Fig.7-12 Height adjusting of cleaning brush****7.4.4.2 Seal Check Conveyor**

Remove and clean the brush once a week or when the brush is extremely dirty.

1. Turn off the main power of equipment.
2. Check that the conveyor belt does not operate and open the cover.

**Fig.7-13 Open the cover**

3. Remove the cleaning brush.
4. Use neutral detergent to wash off the dirt.
5. Naturally dry the cleaning brush and attach it. Check that the tip of cleaning brush touches the belt. If not, adjust the height of attachment.
6. Close the cover.

**Fig.7-14 Remove the cleaning brush**

7.4.5 Cleaning of Conditioning Brush

Disassemble and clean the conditioning brush when the brush is extremely dirty.

1. Turn off the main power of equipment.
2. Check that the conveyor belt does not operate and loosen the screws securing the cover of brush conditioner.
3. Slide and remove the cover.



Fig.7-15 Remove the cover

4. Remove the screws securing the conditioning brush.
5. Pull out the conditioning brush. A conditioning brush consists of a pair of half piece.



Fig.7-16 Remove the conditioning brush

6. Use neutral detergent to wash off the dirt.
7. Naturally dry the conditioning brush and tighten the screws to secure.
8. Attach the cover of brush conditioner and tighten the securing screws.

7.4.6 Stabilizer Unit (option)

Clean the Stabilizer Unit after the daily operation.

1. Use pressurized air to blow off all dusts and product flakes.



Fig.7-17 Stabilizer unit

2. Unscrew the black knobs and pull out the brushes (tools are not required).
3. Use a neutral detergent to wash down the brushes.

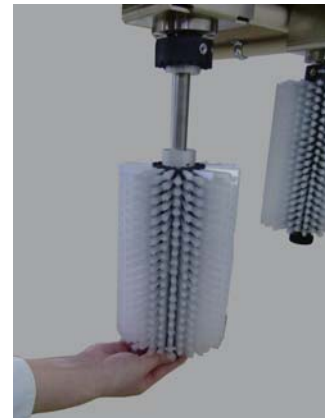


Fig.7-18 Unscrew the knobs

4. Slide back the brushes and match the jagged edge on end of the shafts. Lock down the brushes with black knobs.



Fig.7-19 Reattach the unit